

1 House Price Prediction and Forecasting: Results

1.1 ZIP×Month TAB-MTL Quantile Model

We train a tabular multi-task learning (TAB-MTL) model that operates on a ZIP×Month index, augmented with local market and hedonic features. The model jointly predicts one- and two-month-ahead prices and returns conditional quantiles, which we use to form prediction intervals (PIs). Our primary error metrics are mean absolute error (MAE), coefficient of determination (R^2), weighted absolute percentage error (WAPE), median absolute percentage error (MdAPE), and the fraction of predictions whose absolute percentage error is within $\pm 10\%$ of the truth. For uncertainty calibration, we report PI coverage and the relative PI width.

To avoid over-confident but uninformative predictions, we apply a *confidence gating* rule: predictions whose relative PI width exceeds a fixed threshold are suppressed and excluded from the evaluation set. We also run a rolling backtest with 60-day holdout windows to verify temporal stability of the model.

Table 1: Main holdout performance (last 1–2 months) for the ZIP×Month TAB-MTL quantile model.

Metric	H1 (1-month ahead)	H2 (2-month ahead)
MAE	31.73	31.27
R^2	0.9971	0.9960
WAPE	0.0424	0.0428
MdAPE	0.0311	0.0294
Pct $ \text{err} \leq 10\%$	0.8093	0.8232
PI cover	0.9803	0.9774
Rel. width	0.8913	0.8500

Interpretation. Both horizons achieve extremely strong fit on the fixed holdout, with $R^2 \approx 0.996$ – 0.997 and WAPE around 4%. Roughly 81–82% of predictions fall within $\pm 10\%$ of the true price, indicating high point-forecast accuracy. H2 (two-month ahead) is slightly sharper, with lower MdAPE and narrower intervals (smaller relative width), while maintaining excellent coverage.

Table 2: Suppression / confidence gating statistics on the main holdout for the ZIP×Month TAB-MTL quantile model.

Horizon	Suppressed predictions	Total evaluated	Suppression rate
H1 (1-month ahead)	1038	1369	0.7582
H2 (2-month ahead)	929	1369	0.6786

Interpretation. The gating rule is intentionally conservative: about 76% of H1 and 68% of H2 candidate predictions are suppressed because their intervals are judged too wide. The metrics in Table 1 therefore reflect the high-confidence subset of ZIP×Month pairs where the model determines that the local data manifold is sufficiently well supported.

Table 3: Rolling 60-day backtest results (3 folds, holdout windows only) for the ZIP×Month TAB-MTL quantile model.

Fold	Horizon	MAE	R ²	WAPE	MdAPE	Pct err ≤ 10%	PI cover	Rel. width
1	H1	84.83	0.9738	0.0993	0.0466	0.6584	0.9096	0.9812
1	H2	71.82	0.9837	0.0872	0.0459	0.6744	0.9671	0.9317
2	H1	39.99	0.9977	0.0474	0.0411	0.7529	0.9780	0.9292
2	H2	28.76	0.9976	0.0398	0.0341	0.8286	0.9725	0.8809
3	H1	41.54	0.9932	0.0555	0.0279	0.7889	0.9737	0.7786
3	H2	28.03	0.9978	0.0383	0.0335	0.8028	0.9204	0.6639

Interpretation. Across all three rolling folds, the model maintains high R^2 and low WAPE, confirming that performance is not confined to a single lucky holdout split. H2 consistently delivers lower MAE and tighter intervals than H1 while keeping PI coverage above roughly 92%. H1 is slightly more conservative—intervals are wider and coverage is higher—which can be desirable in risk-sensitive pricing scenarios.

1.2 Latent Manifold Model

Our second algorithm is a Latent Manifold Model based on Local Maximum Entropy (LME). Instead of learning a global regression function, LME constructs a local model around each query property using nearby comparables in the joint feature space of price-per-square-foot (PPSQFT), structural attributes, and location. The local model uses a soft-neighborhood weighting scheme and locally linear regression to approximate the latent price manifold.

We evaluate this model on the PPSQFT task using the same error thresholds reported in the original paper. For a fair comparison, we focus on models that explicitly use location features (“With Loc”) and report the fraction of test properties whose PPSQFT prediction is within 5% or 15% of the ground truth.

Table 4: Test accuracy within price error thresholds (PPSQFT). “With Loc” corresponds to models that use location features. Paper numbers are taken from the reported table; the bottom row shows our implementation.

Algorithm	With Loc	
	< 5%	< 15%
Nearest Neighbor (paper)	27.51	68.31
Neural Network (paper)	27.43	70.10
LME (<i>kernel</i>) (paper)	29.39	71.70
LME (<i>llr</i>) (paper)	29.67	72.06
S - LME (<i>llr</i>) (paper)	29.69	72.15
Our LME (this work)	18.10	50.62

Interpretation. Our current LME implementation under-performs the best paper models on this strict PPSQFT benchmark, especially at the 15% threshold. This gap highlights the sensitivity of LME to the exact choice of distance metric, neighborhood size, and regularization in the local maximum-entropy optimization, and motivates further tuning and feature engineering on this task.

Table 5: Our Latent Manifold model: Train vs Test metrics (PPSQFT).

	< 5%	< 10%	< 15%	Median abs rel
Train	25.54%	49.10%	67.54%	0.1022
Test	18.10%	35.45%	50.62%	0.1479

Interpretation. The train–test gap is modest but noticeable: all three accuracy thresholds drop on the held-out set and the median absolute relative error increases from 0.10 to about 0.15. This suggests that our current LME configuration captures much of the local structure of the price manifold but still overfits to some degree, particularly in sparsely sampled regions of feature space.